2 Jan 2020

Problems

1) Given i) set of locations ii) pairwise distances
   iii) src location iv) dest. location
   Find: Shortest path between src & dest.

2) Given as in 1) i) set of locations, ii) pairwise dist
   iii) src-cum dest.
   Find: A shortest tour. Tour - visit all locations.

3) Given: i) any arbitrary program ii) input to the prog.

Find: Whether prog. terminates (or runs indefinitely)

4) Given: C program
Find: All tokens are valid.

5) Given: C prog. with valid tokens
Find: Legal C program or not.
(Syntactically correct)

Abstract devices

⇒ What are the simplest computational devices?

3), 4) 5) – Decision probs – Yes / No, 1 / 0, True / False

D_4 : solves all 4) problems

D_5 : " " 5) "

→ $D_{1,2}$: " " 1), 2) " ← $D_{1,2} \equiv$ Gen. purpose computer

D_3 : not solvable – cannot have alg to solve 3)
↳ does not exist.

A) Computability questions – $D_4, D_5, D_{1,2}$ – Formally define study prop.
– Why D_3 does not exist?

B) 1), 2), 4), 5) are solvable. How tractable? ✓
↳ How much time, space as a fn of input size?

Complexity questions.

$O(n), O(n^2), \ldots$

tractable [1) $O(n^2)$, $n \equiv$ no of locations
4), 5) $O(n)$ $n \equiv$ length of program

$\geq O(2^n)$ [2) TSP – Travelling salesperson problem. – $O(n!)$, $O(n^2 2^n)$
$n \equiv$ no. of locations.

Polynomial problems – P
Exponential " – NP

$P \stackrel{?}{=} NP$ open problem
$P \neq NP$

Tractable	Intractable	Unsolvable
1), 4), 5)	2)	3)

→ $O(n^{100})$ – $O(2^n)$

All alg. $O(n^m)$ m is small < 10

What is computation?

1), 2) string → string Yes/No, 1/0
locs, dist, src loc, dest → seq. locs
fn : strings → strings

Σ : Alphabet of symbols

Σ^* : All possible strings with alphabet Σ.

$\Sigma = \{0, 1\}$

$\Sigma^* = \{0, 1, 00, 01, 10, 11, 000, 001, 010, \ldots\}$

→ fn : $\Sigma_i^* \rightarrow O_s$ Y/N, 1/0, True/False

Decision problems prob 4)

$\Sigma = \{a-z, A-Z, +, -, /, *, ., 0-9, ';', ','\}$

$\Sigma = \{0-9\}$ $\Sigma^* \equiv$ all poss. numbers. $\mathbb{N}_0$

Given $s \in \Sigma^*$ is it even or not.

Finite automaton:

$\Sigma = \{0-9\}$

125768

~~1783~~

$q_0 \xrightarrow{1} q_0$

$q_0 \xrightarrow{2} q_1$

$q_1 \xrightarrow{5} q_0$

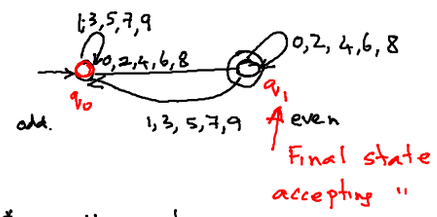
$q_0 \xrightarrow{7} q_0$

$q_0 \xrightarrow{6} q_1$

$q_1 \xrightarrow{8} q_1$

s, q_0

F_{even}



$\Sigma^* \equiv$ all numbers

$L \subset \Sigma^*$, $L \equiv$ all even nos

F_{even} - Finite automaton which accepts L

if $s \in L$, F_{even} accepts s., $\forall s \in L$, F_{even} accepts s.

$L \equiv$ Langs accepted by FA - regular langs.

Regular langs
- FA — acceptance device
- Reg. Expns.
- Reg. Grammars — generative device.

Finite automaton - Deterministic

$FA = (\Sigma, Q, \delta, q_0, F)$

- Σ: Finite Alphabet
- Q: set of states (Finite)
- δ: transition fn
- $q_0 \in Q$ - initial state
- $F \subseteq Q$ set of final states.

$\Sigma = \{\sigma_1, \dots \sigma_m\}$

$\delta : Q \times \Sigma \to Q$ - deterministic.
- trans on every symbol
- exactly one trans. on a symbol

DFA $\equiv$ Deterministic Fin. Automaton.

	σ_1	σ_2	σ_m
q_0			
q_2			
$\vdots$			
q_n			

$\equiv \delta$

Examples: $\Sigma = \{0, 1\}$

1) Identify all strings that end in 0.

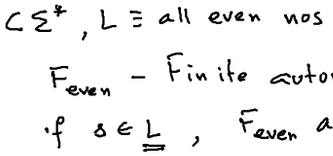
2) " " " " " " 1.

3) " " " " begin and end with same symbol.

7 Jan.

1)

last sym 0.



$q_0 \equiv$ last symbol 1

$q_1 \equiv$ " " 0

$\{0, 00, \dots, 10, \dots\}$

will not have 1, 11, 01, ...

$q_0 \xrightarrow{1} q_0$

$q_0 \xrightarrow{1} q_0$

$q_0 \xrightarrow{0} q_1$

$q_1 \xrightarrow{1} q_0$

1010

1101

$q_0 \xrightarrow{1} q_0$

$q_0 \xrightarrow{0} q_1$

$q_1 \xrightarrow{1} q_0$

$q_0 \to q_1$

$q_0 \xrightarrow{1} q_0 \xrightarrow{0} q_1 \xrightarrow{1} q_0 \xrightarrow{0} q_1$

$q_0 \xrightarrow{1} q_0 \xrightarrow{1} q_0 \xrightarrow{0} q_1 \xrightarrow{1} q_0$

Defn: string $w \in \Sigma^*$ is accepted by DFA M iff $\delta(q_0, w = \sigma_1\sigma_2 \cdots \sigma_k)$

$M = (Q, \Sigma, \delta, q_0, F)$

$\Sigma = \{0,1\}$

$\Sigma^+ = \{0,1,00,10, \ldots\}$

$\Sigma^* = \{\varepsilon, 0, 1, 00, \ldots\}$

$\delta(\delta \ldots (\delta(q_0, \sigma_1), \sigma_2) \ldots, \sigma_k) \in F$

$\delta : Q \times \Sigma \to Q \to$ trans on a symbol

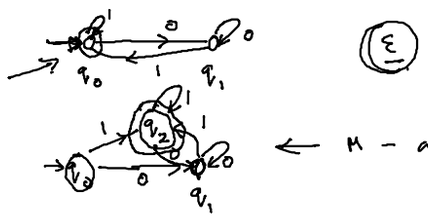
$\hat{\delta} : Q \times \Sigma^* \to Q \to$ " " " string

w is accepted by DFA M iff $\hat{\delta}(q_0, w) \in F$

Defn: Lang. accepted by DFA M

$L(M) = \{w \in \Sigma^* \mid M \text{ accepts } w\}$

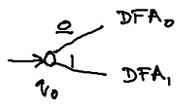
2)



ε

← M – accepts all strings ending in 1.

3)



DFA_0 all strings end in 0

DFA_1 all " " " 1

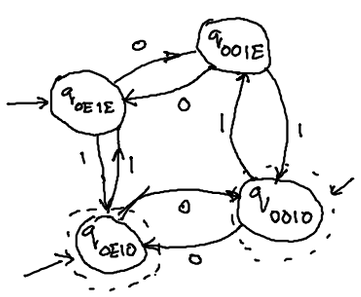
$\Sigma = \{0,1\}$ 4) no of 0s is odd/even, no of 1s is odd/even.

↳ q_{0E1E} — 0s even & 1s even

q_{0E1O} — " & 1s odd

q_{0O1E} — 0s odd & 1s even

q_{0O1O} — " & 1s odd



even 0s odd 1s

odd 0s odd 1s

DFA — a) trans on each symbol in Σ for each state.

Deterministic

Prob: All strings over $\Sigma = \{0,1\}$ s.t. the third last symbol is a 1.

$\{00100, 100, 111, 0100, \ldots\}$

Non-deterministic automaton (NFA)

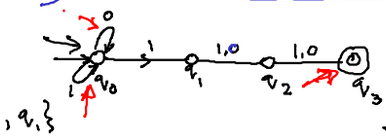
↳ $\boxed{\text{Guess and Check}}$

Non-det ⇒
* In a state 0 or more transitions on a sym (Σ).
* Allows trans on ε

100, 101, 110, 111

$q_0 = 2$ trans on 1

$q_3 = 0$ " .



δ: $q_0 \xrightarrow{1} \{q_0, q_1\}$

$q_3 \xrightarrow{0/1} \{\}$

0101

$\Rightarrow q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_1 \xrightarrow{0} q_2 \xrightarrow{1} q_3$ F

$\xrightarrow{1} q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_1$ $\{q_3, q_1, q_0\}$

$\to q_0$

$q_0 \xrightarrow{0} q_0 \to q_1$

$\to q_0$

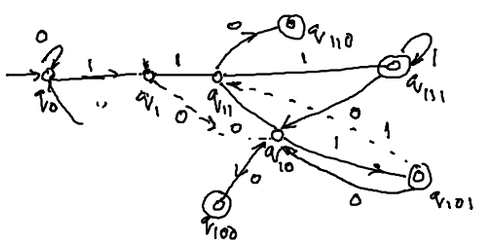
001

$q_0 \xrightarrow{0} q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_1$ $\{q_1, q_0\}$ 001

$\to q_0$

Same lang $L = \{w \in \Sigma^* \mid 3^{rd}$ last sym is $1\}$, $\Sigma = \{0,1\}$.

100, 101, 110, 111



q_1
q_0,
q_{10},
q_{11}
q_{100}
q_{101}
q_{110}
q_{111}

110
100
110
010
011

YES

YES

Are NFAs $\equiv$ DFAs? Are they equally powerful?

For every lang recog by an NFA does there exist a DFA which accepts the same lang.?
(recog: accepted)

NFA $= (Q, \Sigma, \delta, q_0, F)$ same as DFA.

Only difference in δ

NFA: $\delta : Q \times \Sigma \longrightarrow Q' \subseteq Q$ (ϕ)

$\longrightarrow \mathcal{P}(Q) \equiv$ Power set Q.

DFA: $\delta : Q \times \Sigma \to Q$

Defn: string w is accepted by NFA M iff $\hat{\delta}(q_0, w) \cap F \neq \phi$.

$L(M) = \{w \in \Sigma^* \mid w$ is accepted by $M\}$.

Regular langs $\equiv$ All langs that are accepted by DFA/NFA $\equiv$ FA

Token validity

C-prog $\longrightarrow$ converted into tokens (valid)
lexical analysis
NFA

Assgn 1

Mobile payments
- wallet (Customer's) - app. (PayTM, PhonePe, GooglePay, OlaMoney)
- Service provider
- Vendor
- Bank/credit/Debit card (change your wallet)

Pay, cancel, recharge,

First main result

$\Rightarrow$ Theorem 1: NFA $\equiv$ DFA ✓

A lang $L \subseteq \Sigma^*$ is recognized/accepted by an NFA iff it is accepted by a DFA.

if L is accepted by DFA then it is accepted by an NFA.

Trivial: Because every DFA is also an NFA.

if L is accepted by an NFA then it is accepted by a DFA.

$\hookrightarrow$ subset construction,

Read chp 0 of Sipser's book.

Quiz on chp 0 of Sipser's book on 18th or 25th

9 Jan

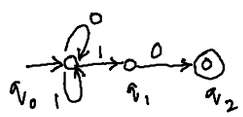
DFA
- Exactly 1-trans on a sym in Σ
- a trans in every single in Σ

NFA - Both above conds relaxed

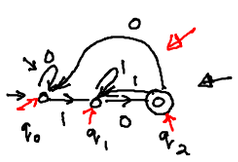
$\rightarrow \Rightarrow$ · 0 or more trans on a sym
· trans without a sym
ε- trans

$\Sigma = \{0,1\}$ $L = \{w \in \Sigma^* \mid$ last 2 chars are $10\}$ | Guess & Check

$\Rightarrow$



$\boxed{0110}$ accepted.



$q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_0 \xrightarrow{1} q_1 \xrightarrow{0} q_2$ $\quad q_2, q_0$

$\vdots \xrightarrow{1} q_1$ X

0110 ✓

$\rightarrow$ 0101 not accepted.

$q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_1$ $\quad q_1, q_0$

$\vdots \xrightarrow{1} q_1 \xrightarrow{0} q_2$ X $\xrightarrow{1} q_0$

$q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_1 \xrightarrow{0} q_2 \xrightarrow{1} q_1$

0101 X

$q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_1 \xrightarrow{1} q_1 \xrightarrow{0} q_2$

$\delta(q_0, 1) = \{q_0, q_1\}$

$\delta : Q \times \Sigma \rightarrow Q' \subseteq Q$

✓ $L' = \{w \in \Sigma^* \mid 3^{rd}$ last sym is $1\}$

L' DFA states > NFA states.

L " = "

NFA $\equiv$ DFA.

if $L(M)$, M is an NFA then there exists a DFA M'
$L(M') = L(M)$.

Reg Langs $\equiv$ Langs accepted by some NFA | DFA

$\Sigma = \{0, 1, \ldots 9, +, -, \cdot\}$

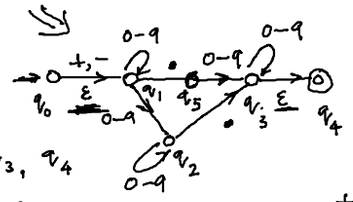
$L = \underset{\text{nothing}}{\pm}$ seq digits1. seq digits2

constraint at least seq digits 1 or seq digits 2 must be present

Examples: $+0.123$ $\quad 312.123$ $\quad -.123$
$-123.$

$\cdot S2$

$S1\cdot$



$\boxed{\frac{+}{\varepsilon}\ \underline{S1}\cdot\underline{S2}}$

$S1, S2$ — at least one

$\frac{+}{\varepsilon}\ \cdot S2$

$S1\cdot$

$\underline{S1\cdot}$ $q_0\ q_1\ q_2\ q_3,\ q_4$

$\cdot S2$ $q_0\ q_1\ q_5\ q_3\ q_4$

$+1\cdot 2$

ε− transition

$Q' = \{q_0, q_3\}$

$\varepsilon\text{-}Q' = \{q_0, q_3, q_1, q_4\}$

$q_0 \xrightarrow{\varepsilon} q_1 \xrightarrow{1} q_1 \xrightarrow{\cdot} q_3 \xrightarrow{2} q_3 \xrightarrow{\varepsilon} q_4$

$\vdots \xrightarrow{1} q_2 \xrightarrow{\cdot} q_3$

$-\cdot 21$

$q_0 \xrightarrow{\varepsilon} q_1 \xrightarrow{\cdot} q_3 \xrightarrow{2} q_3 \xrightarrow{1} q_3 \xrightarrow{\varepsilon} q_4$

$q_0 \xrightarrow{\varepsilon} q_1 \xrightarrow{2} q_1 \xrightarrow{1} q_1 \xrightarrow{\cdot} q_3 \to q_4$ $21\cdot$

$\vdots \to q_2 \times$

NFA − no trans on any sym
mult. " " a "
ε trans

If L is accepted by an NFA it is accepted by a DFA.

Subset constrn.

NFA $M = (Q, \Sigma, \delta, q_0, F)$

$\mathcal{P}(Q) \equiv \{Q' \mid Q' \subset Q\}$ — power set of Q.

DFA $M' = (\mathcal{P}(Q), \Sigma, \delta', \{q_0\}, F')$

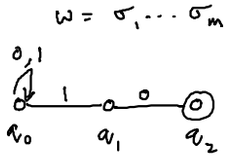
$F' =$ all $Q' \subseteq Q$ s.t. $Q' \cap F \neq \phi$

$\delta'(\underbrace{\{q_{i_1} \cdots q_{i_m}\}}_{Q'}, \sigma) = \bigcup_{q \in Q'} \delta(q, \sigma)$

$\varepsilon\text{-}Q'$ = set of all states reachable state in Q' via ε- trans.

Claim : $\hat{\delta}(q_0, \underline{w}) = Q^* = \hat{\delta}'(\{q_0\}, w)$

$w = \sigma_1 \cdots \sigma_m$



− 10 − last 2 symbols.

NFA = $(\{q_0, q_1, q_2\}, \delta, q_0, q_2)$

DFA = $(\mathcal{P}(\{q_0, q_1, q_2\}, \delta', \{q_0\}, F')$

$\{q_0\}\ \{q_1\}\ \{q_2\}$

$\{q_0, q_1\}\ \{q_0, q_2\}\ \{q_1, q_2\}$

$\{q_0, q_1, q_2\}$

$q_0 \xrightarrow{1} \{q_0, q_1\}$

$q_0 \xrightarrow{0} q_0$

$q_1 \xrightarrow{0} q_2$

$\{q_0\} \xrightarrow{0} \{q_0\}$

$\{q_0\} \xrightarrow{1} \{q_0, q_1\}$

$\{q_0, q_1\} \xrightarrow{0} \{q_0, q_2\}$, $\{q_0, q_1\} \xrightarrow{1} \{q_0, q_1\}$

$\{q_0, q_2\} \xrightarrow{0} \{q_0\}$

$\{q_0, q_2\} \xrightarrow{1} \{q_1\}$